



HUST

ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.



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FUNDAMENTALS OF OPTIMIZATION

Introduction

ONE LOVE. ONE FUTURE.

INTRODUCTION TO OPTIMIZATION PROBLEMS

- Maximize or minimize some function relative to some set (range of choices)
- The function represents the quality of the choice, indicating which is the “best”
- Example
 - A shipper need to find the shortest route to deliver packages to customers 1, 2, ..., N

0	3	1	6
3	0	2	4
1	2	0	5
6	4	5	0

0	3	4	2	5	6	8	7
3	0	3	6	7	2	1	6
4	3	0	4	7	1	1	9
2	6	4	0	2	8	3	4
5	7	7	2	0	6	5	1
6	8	1	8	6	0	9	3
8	1	1	3	5	9	0	2
7	6	9	4	1	3	2	0

INTRODUCTION TO OPTIMIZATION PROBLEMS

- $x \in R^n$: vector of decision variables x_j for $j = 1, 2, \dots, n$
- $f: R^n \rightarrow R$ is the objective function
- $g_i: R^n \rightarrow R$ is the constraint function defining restriction on x , $i = 1, 2, \dots, m$

minimize $f(x)$ over $x = (x_1, x_2, \dots, x_n) \in X \subset R^n$ satisfying a property P :

$$g_i(x) \leq b_i, i = 1, 2, \dots, s$$

$$g_i(x) = d_i, i = s + 1, 2, \dots, m$$

INTRODUCTION TO OPTIMIZATION PROBLEMS

- Example

$$\begin{aligned}\min f(x) &= 3x_1 - 5x_2 + 10x_3 \\ x_1 + x_2 + x_3 &\leq 10 \\ 2x_1 + 4x_2 - 5x_3 &= 9 \\ x_1, x_2 &\in \mathbb{R}^+, x_3 \in \mathbb{Z}\end{aligned}\quad (\text{Linear Program})$$

$$\begin{aligned}\min f(x) &= 4x_1^2 + 3x_2^2 - 7x_1 x_3 \\ x_1 + x_2^3 + 4x_3 &\leq 10 \\ 2x_1^2 + 4x_2 - 5x_3 &= 7 \\ x_1, x_2 &\in \mathbb{R}^+, x_3 \in \mathbb{Z}\end{aligned}\quad (\text{Nonlinear Program})$$

INTRODUCTION TO OPTIMIZATION PROBLEMS

- General optimization problems
 - Very difficult to solve
- Some special cases
 - Linear programming
 - Least square problem
 - Some shortest path problems on networks
 - Etc.

INTRODUCTION TO OPTIMIZATION PROBLEMS

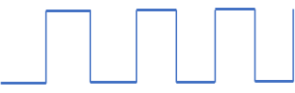
- Classification
 - Linear Programming (LP): f and g_i are linear
 - Nonlinear Programming (NLP): some function f, g_i are nonlinear
 - Continuous optimization: f and g_i are continuous on an open set containing X , X is closed and convex
 - Integer Programming (IP): $X \subseteq \{0,1\}^n$ or $X \subseteq \mathbb{Z}^n$
 - Constrained optimization: $m > 0, X \subset \mathbb{R}^n$
 - Unconstrained optimization: $m = 0, X = \mathbb{R}^n$

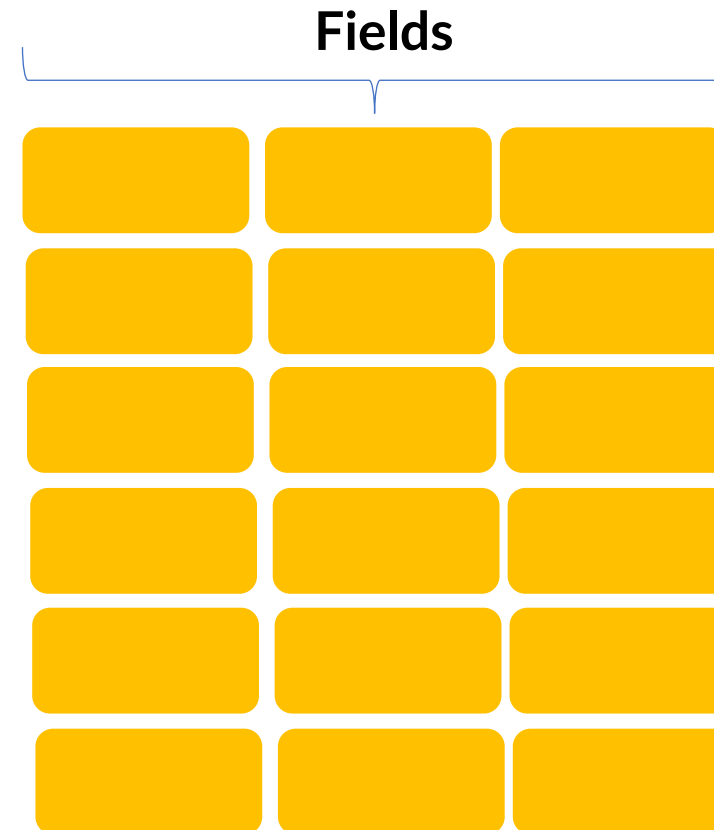
INTRODUCTION TO OPTIMIZATION PROBLEMS

- Applications
 - Production Planning
 - Routing in transportation
 - Scheduling
 - Assignment
 - Packing
 - Time Tabling
 - Network designs
 - Machine learning
 - ...

INTRODUCTION TO OPTIMIZATION PROBLEMS

- Production Planning

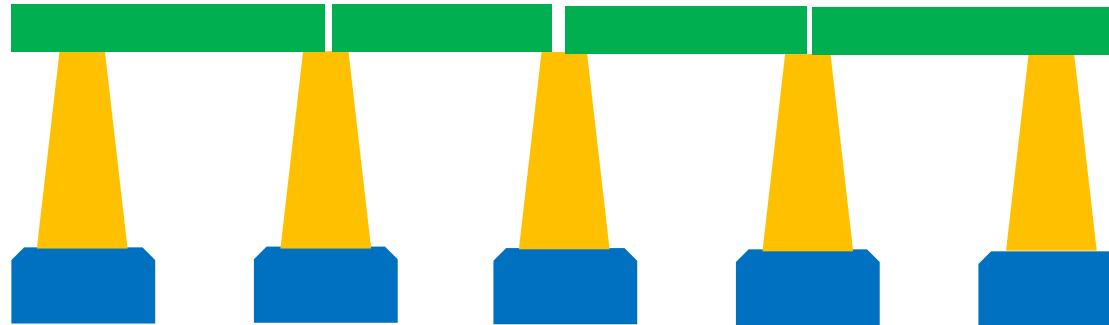
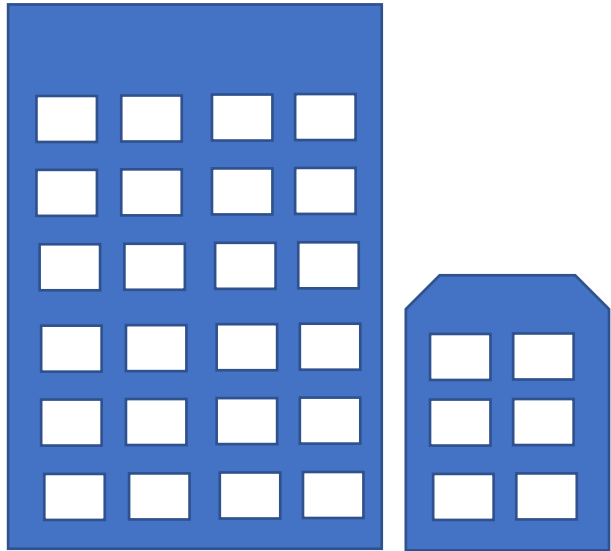
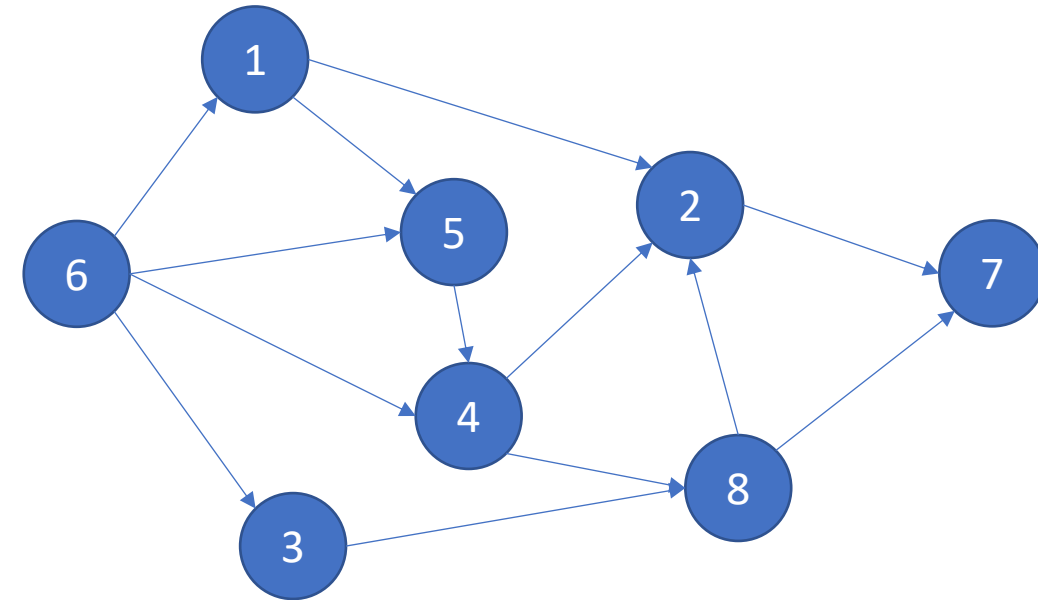
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INTRODUCTION TO OPTIMIZATION PROBLEMS

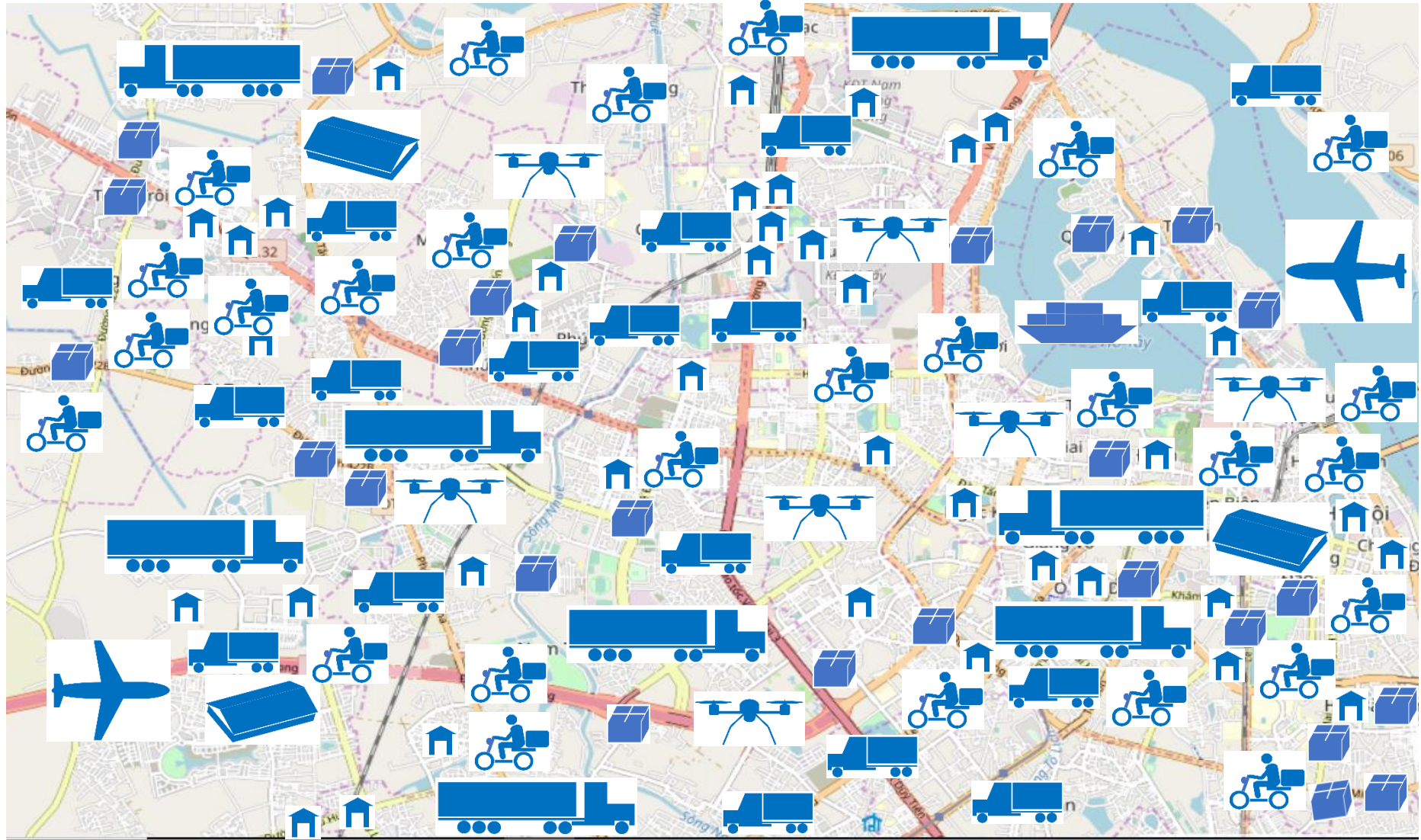
- Construction Planning

Task	Duration	Predecessors
1	30	6
2	20	1,4,8
3	15	6
4	25	5,6
5	20	1,6
6	45	
7	40	2,8
8	30	3,4



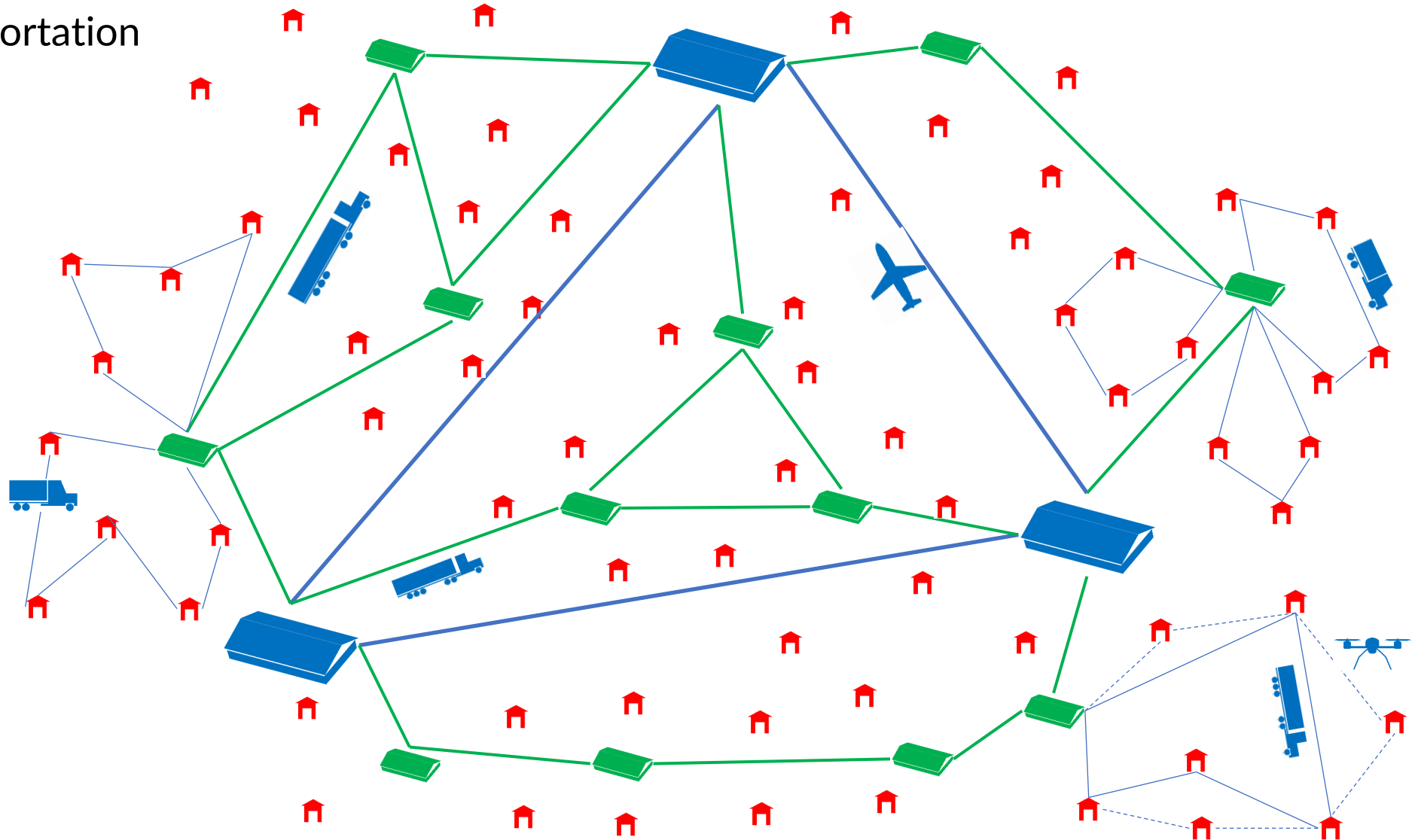
INTRODUCTION TO OPTIMIZATION PROBLEMS

- Routing in transportation & logistics



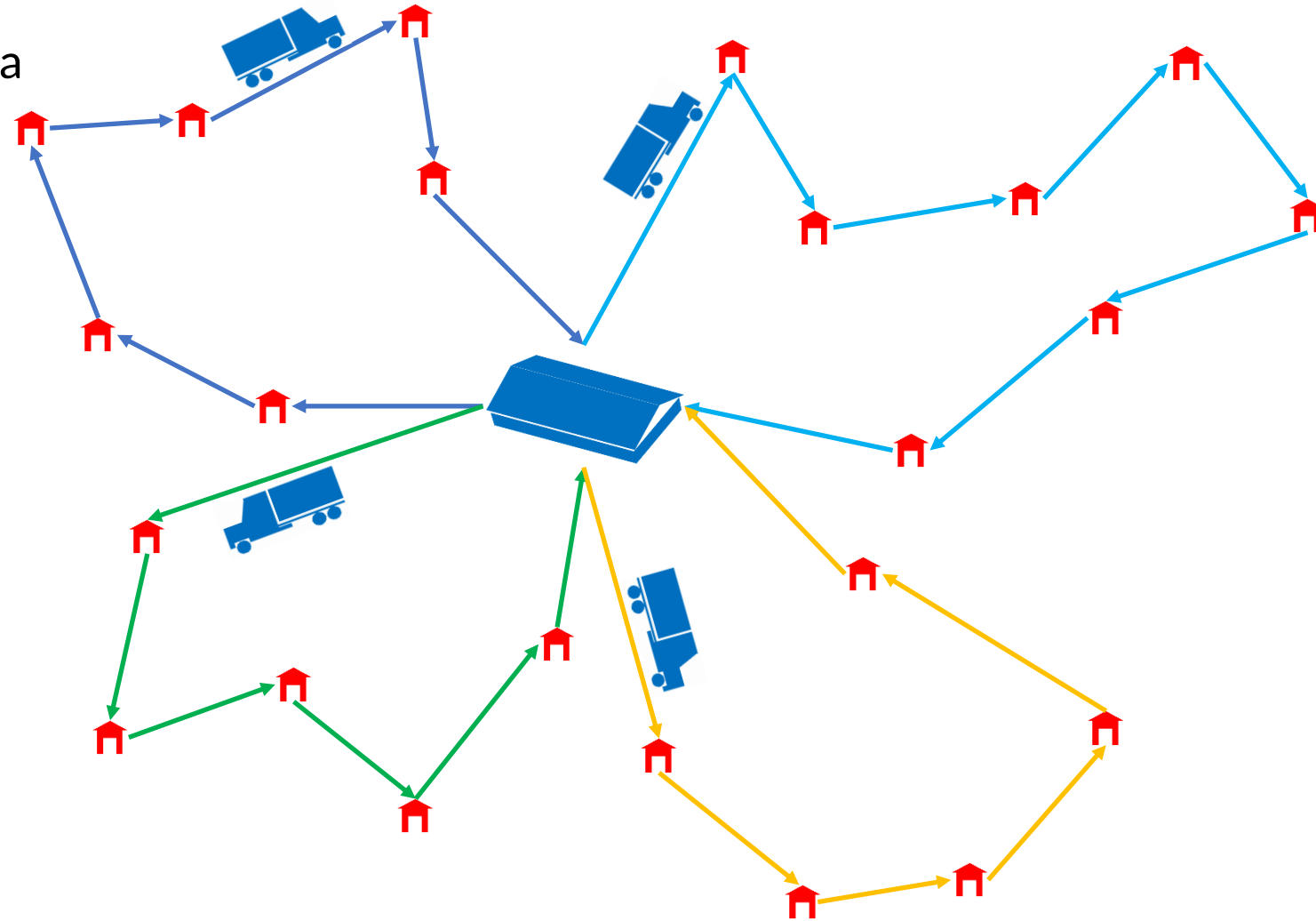
INTRODUCTION TO OPTIMIZATION PROBLEMS

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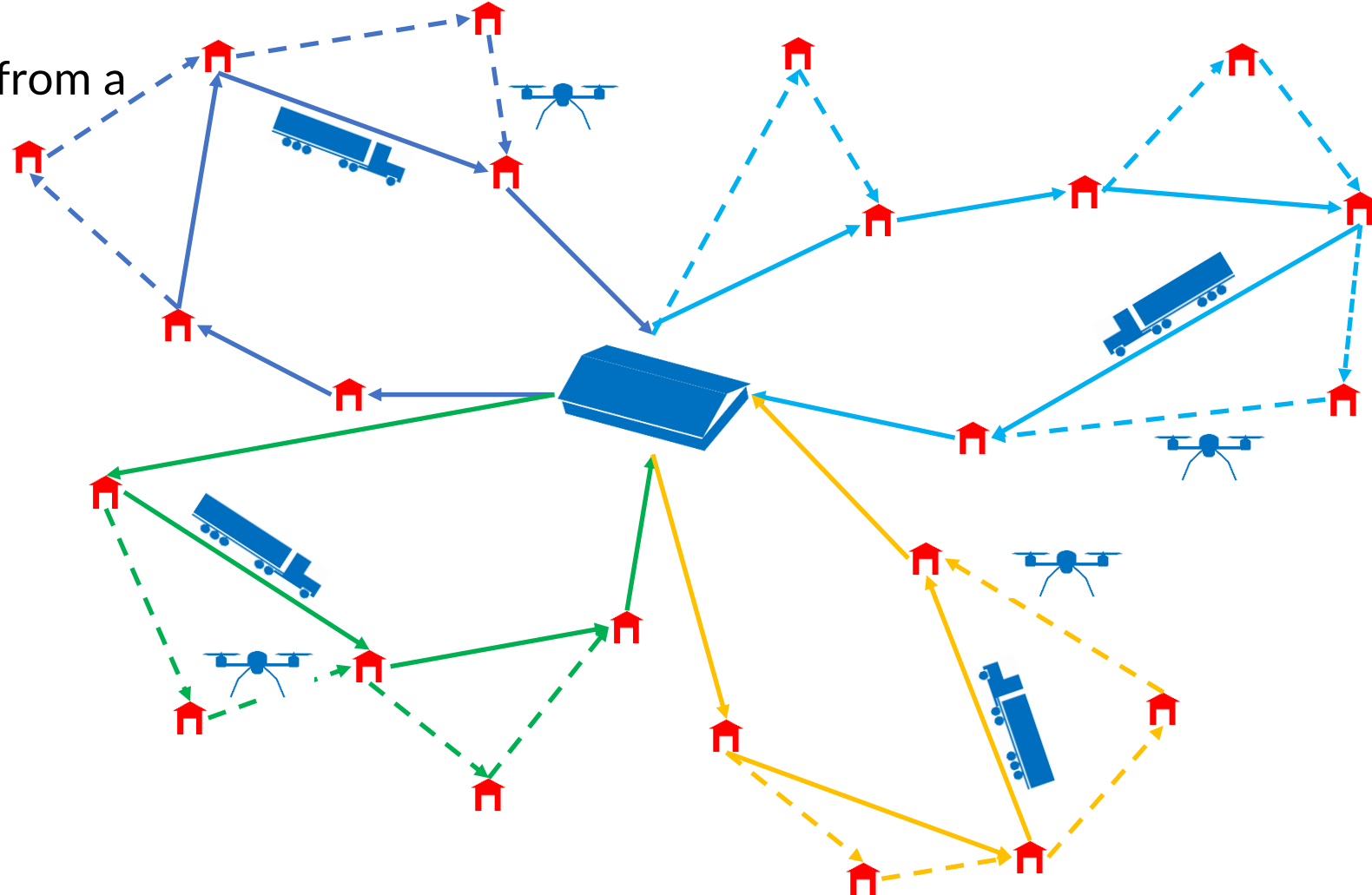
INTRODUCTION TO OPTIMIZATION PROBLEMS

- Routing in transportation & logistics
 - How to make a route plan for delivering goods to customers from a central depot?



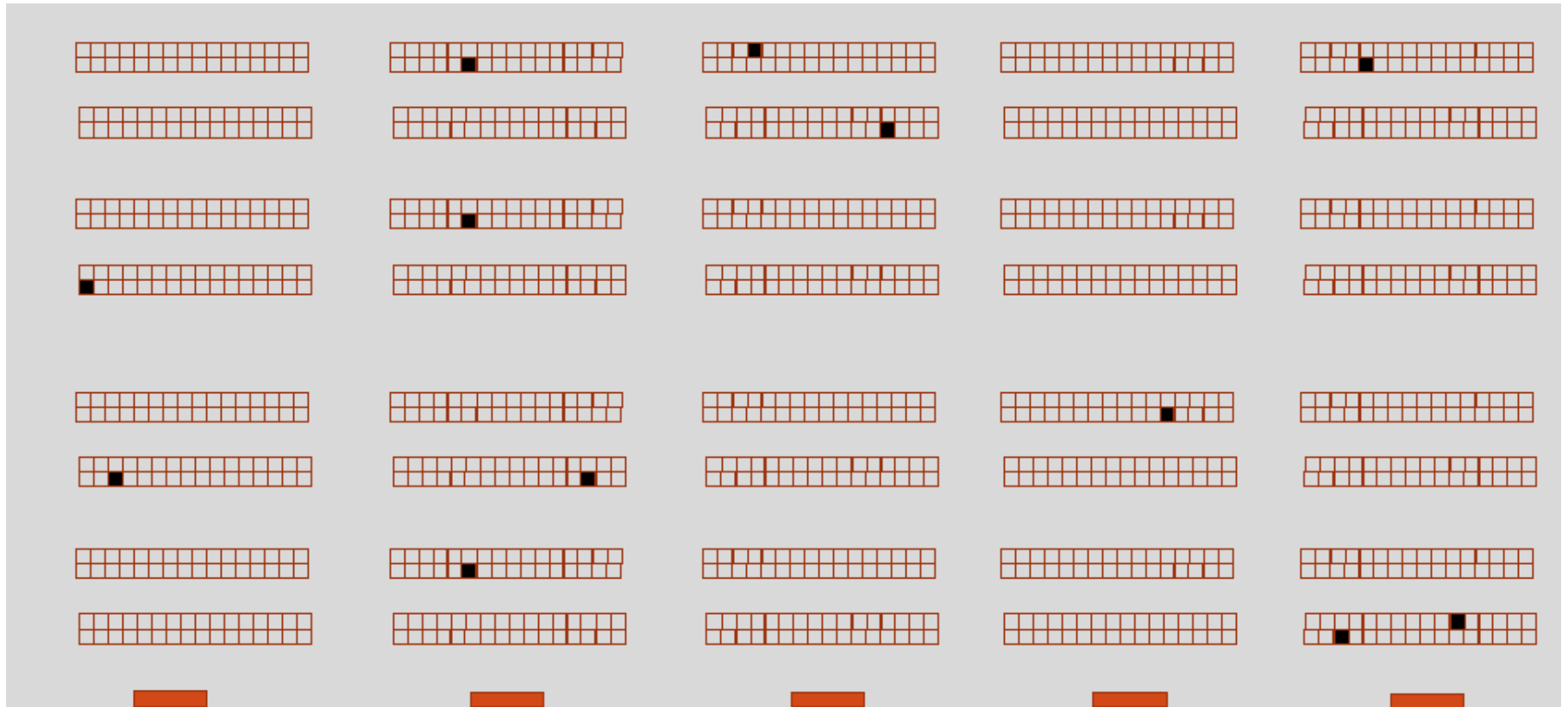
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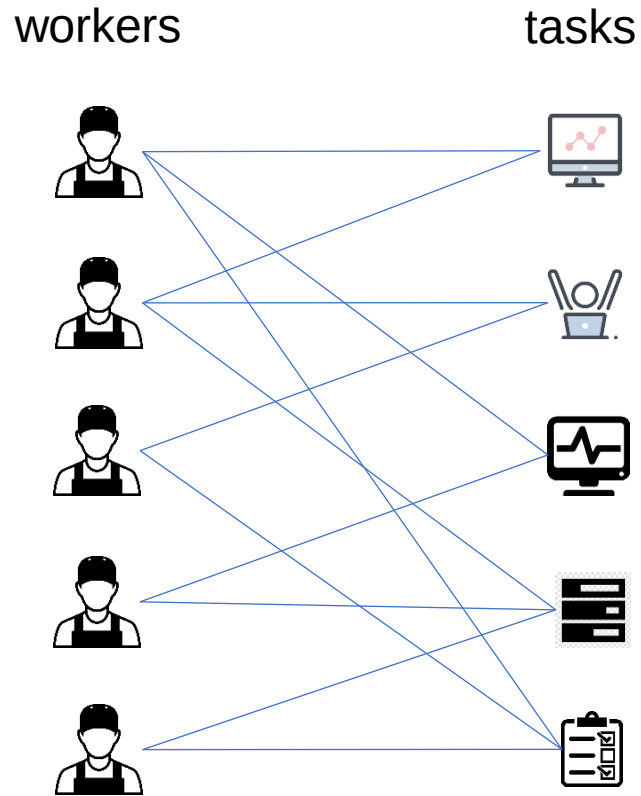
INTRODUCTION TO OPTIMIZATION PROBLEMS

- Routing in transportation & logistics
 - How to make a route plan for picking up items in a very large warehouse?



INTRODUCTION TO OPTIMIZATION PROBLEMS

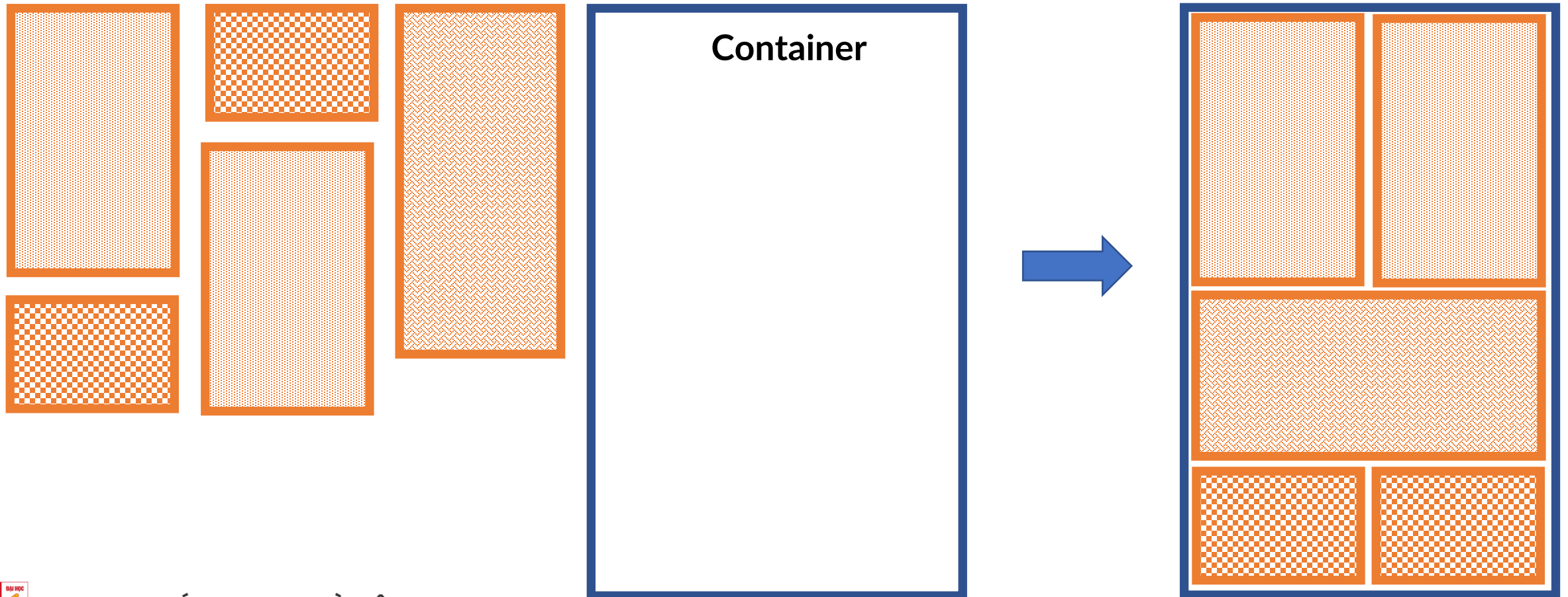
- Assignment
 - How to assign tasks to workers in an optimal way ?



4		6		8
2	6		7	
	5			6
		1	4	
			6	3

INTRODUCTION TO OPTIMIZATION PROBLEMS

- Packing
 - How to arrange items in a container in an optimal ways?



INTRODUCTION TO OPTIMIZATION PROBLEMS

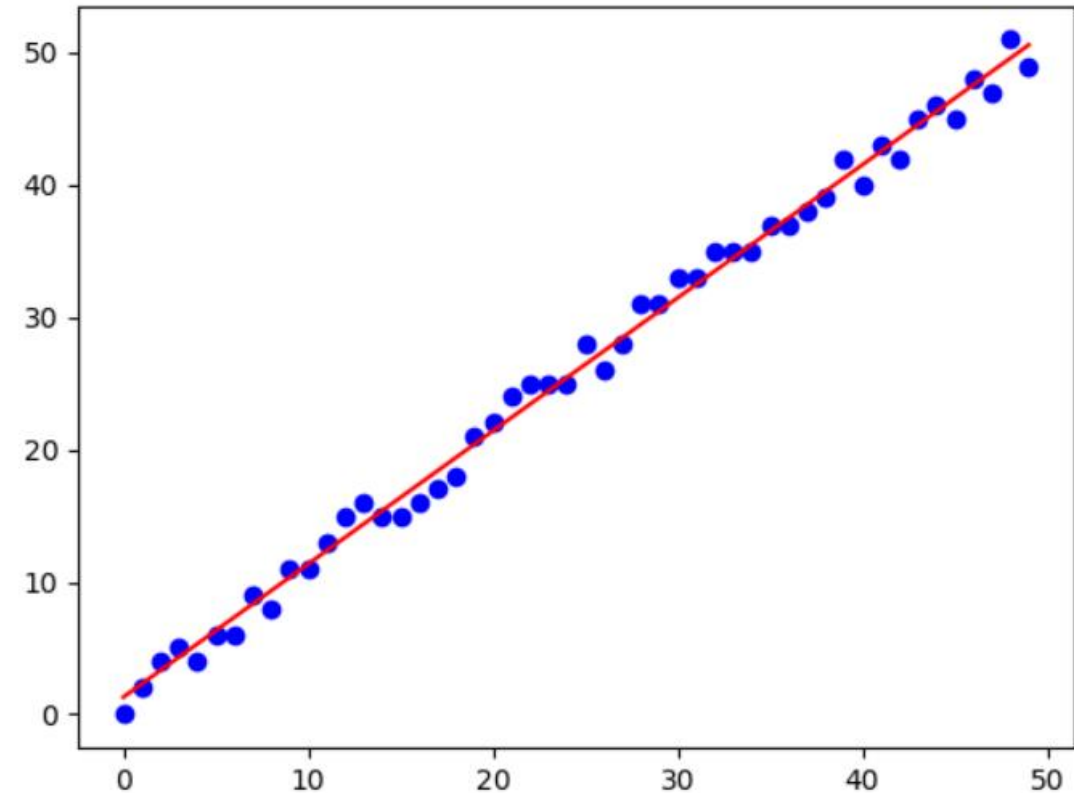
- Time Tabling
 - How to arrange courses into time slots?

Monday	Tuesday	Wednesday	Thursday	Friday
Data structure & Algorithms, TC-305	Python Programming, D9-302	Statistics, B1-203	Technical writing, B1-202	Networkings , B1-404
Fundamenta l of optimization, B1-402			Java advanced, B1-204	
	Machine learning, D6-302	Software engineering, D5-102	Operating systems, D9-101	Image processing, D6-303

INTRODUCTION TO OPTIMIZATION PROBLEMS

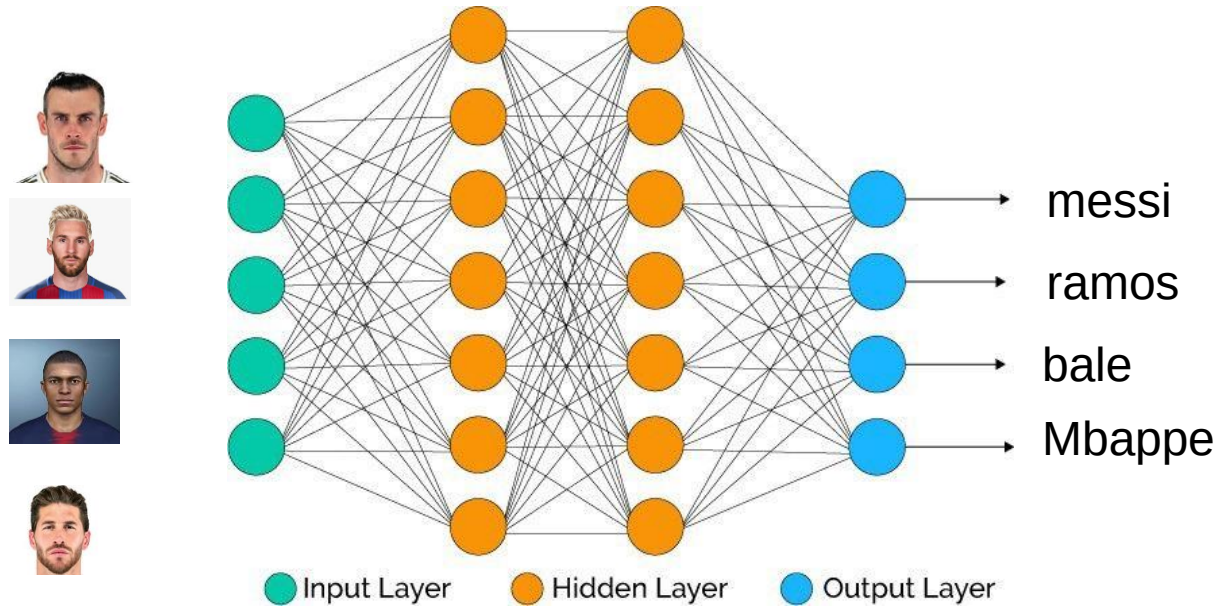
- Machine learning
 - Prediction

X	Y
43	45
44	46
45	45
46	48
47	47
48	51
49	49
50	?



INTRODUCTION TO OPTIMIZATION PROBLEMS

- Machine learning
 - Computer Vision



A graphic on the left side of the slide. It features a dark blue background with a large, stylized circular pattern of red dots. The dots are arranged in concentric, slightly irregular rings, creating a textured, halftone-like effect. In the center of this pattern, the word "HUST" is written in a bold, white, sans-serif font.

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THANK YOU !